

# Zhihan Yin

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## RESEARCH INTERESTS

Multimodal agents; autonomous cross-modal search; vision-language model reasoning; fine-grained visual perception and hallucination; alignment and RL post-training.

## EDUCATION

**Peking University** | Aug 2023 - Present

B.S. Candidate, Intelligence Science and Technology | Cumulative GPA: 3.69 / 4.00

## SELECTED PUBLICATIONS AND MANUSCRIPTS

### MIRA: Reinforcing Multimodal Reasoning via Deceptive Contextual Augmentation

**Zhihan Yin**, Jianxin Liang, Yifeng Yao, Nonghai Zhang, Bingyue Peng, Chunguang Qie, Huishuai Zhang, Dongyan Zhao

*NeurIPS 2026 · Poster*

Reinforcement learning with deceptive visual contexts and group-wise reflection triggers, encouraging multimodal models to revisit perceptual mistakes and recover during reasoning.

### FREAK: A Fine-grained Hallucination Evaluation Benchmark for Advanced MLLMs

**Zhihan Yin**, Jianxin Liang, Yueqian Wang, Yifeng Yao, Huishuai Zhang, Dongyan Zhao

*ICLR 2026*

A benchmark built from photorealistic images with fine-grained counter-commonsense edits, probing whether advanced multimodal models perceive visual details or fall back on familiar priors.

### Veto: A Training-free Framework for Fine-grained Visual Perception via Iterative Negative Filtering

Ruyi Li, **Zhihan Yin**, Tan Yue, Jianxin Liang, Xian-Feng Han, Huishuai Zhang, Dongyan Zhao

*ACM Multimedia 2026*

A training-free perception framework that progressively removes distracting image regions, helping vision-language models locate and verify fine-grained visual evidence.

## RESEARCH AND TEACHING EXPERIENCE

### Research Intern, Monetization GenAI, ByteDance

Apr 2026 - Present

Research on multimodal agents.

### Research Intern, Wangxuan Institute of Computer Technology, Peking University

Mar 2025 - Present

Research on multimodal large language models, visual perception, hallucination evaluation, and reasoning.

### Teaching Assistant, Introduction to Artificial Intelligence, Peking University

Mar 2026 - Present

Course instruction support, assignment design and grading, and student mentoring.

## HONORS AND AWARDS

**Merit Student** (Top 10% at Peking University) | 2025, 2026

**Huirong Li Scholarship** (Top 15% at Peking University) | 2025

**National Encouragement Scholarship** (Top 30% at Peking University) | 2024, 2025

## SKILLS

**Programming:** Python, C++

**Deep learning:** PyTorch, vLLM, veRL

**Research tools:** LaTeX, Adobe Illustrator

**English:** CET-4: 580; CET-6: 570